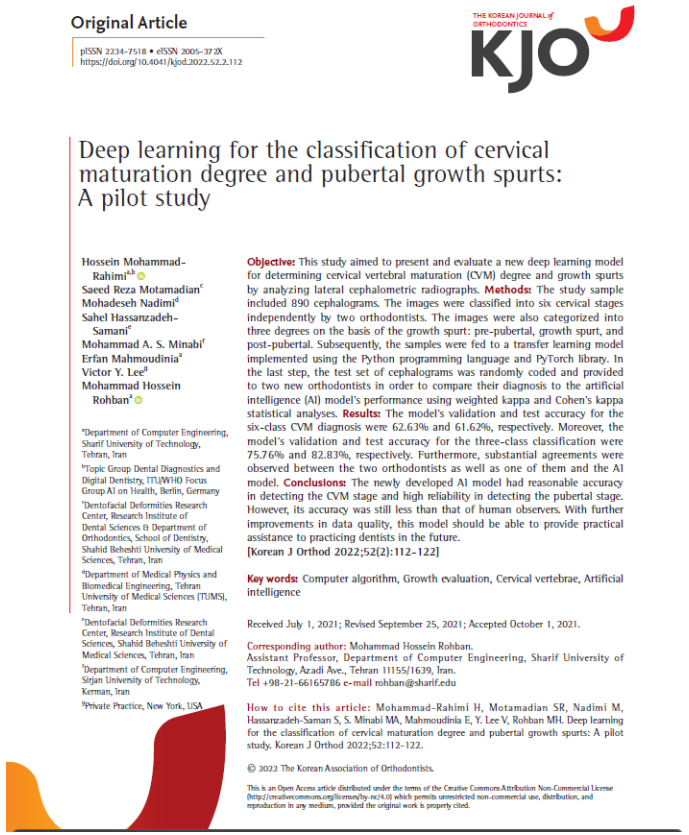
Deep learning for the classification of cervical maturation degree and pubertal growth spurts: A pilot study

- Year: 2022
- Journal Name: Korean Journal of Orthodontics
- Scopus Quartile: Q1 Percentile: 87th
- https://doi.org/10.4041/kjod.2022.52.2.112



Original Article

pISSN 2234-7518 • eISSN 2005-372X
https://doi.org/10.4041/kjod.2022.52.2.112

THE KOREAN JOURNAL of ORTHODONTICS KJO

Deep learning for the classification of cervical maturation degree and pubertal growth spurts: A pilot study

Hossein Mohammad-Rahimi[a,b]
Saeed Reza Motamadian[c]
Mohadeseh Nadimi[d]
Sahel Hassanzadeh-Samani[e]
Mohammad A. S. Minabi[f]
Erfan Mahmoudinia[a]
Victor Y. Lee[g]
Mohammad Hossein Rohban[a]

[a]Department of Computer Engineering, Sharif University of Technology, Tehran, Iran
[b]Topic Group Dental Diagnostics and Digital Dentistry, ITU/WHO Focus Group AI on Health, Berlin, Germany
[c]Dentofacial Deformities Research Center, Research Institute of Dental Sciences & Department of Orthodontics, School of Dentistry, Shahid Beheshti University of Medical Sciences, Tehran, Iran
[d]Department of Medical Physics and Biomedical Engineering, Tehran University of Medical Sciences (TUMS), Tehran, Iran
[e]Dentofacial Deformities Research Center, Research Institute of Dental Sciences, Shahid Beheshti University of Medical Sciences, Tehran, Iran
[f]Department of Computer Engineering, Sirjan University of Technology, Kerman, Iran
[g]Private Practice, New York, USA

Objective: This study aimed to present and evaluate a new deep learning model for determining cervical vertebral maturation (CVM) degree and growth spurts by analyzing lateral cephalometric radiographs. **Methods:** The study sample included 890 cephalograms. The images were classified into six cervical stages independently by two orthodontists. The images were also categorized into three degrees on the basis of the growth spurt: pre-pubertal, growth spurt, and post-pubertal. Subsequently, the samples were fed to a transfer learning model implemented using the Python programming language and PyTorch library. In the last step, the test set of cephalograms was randomly coded and provided to two new orthodontists in order to compare their diagnosis to the artificial intelligence (AI) model's performance using weighted kappa and Cohen's kappa statistical analyses. **Results:** The model's validation and test accuracy for the six-class CVM diagnosis were 62.63% and 61.62%, respectively. Moreover, the model's validation and test accuracy for the three-class classification were 75.76% and 82.83%, respectively. Furthermore, substantial agreements were observed between the two orthodontists as well as one of them and the AI model. **Conclusions:** The newly developed AI model had reasonable accuracy in detecting the CVM stage and high reliability in detecting the pubertal stage. However, its accuracy was still less than that of human observers. With further improvements in data quality, this model should be able to provide practical assistance to practicing dentists in the future.

[Korean J Orthod 2022;52(2):112-122]

Key words: Computer algorithm, Growth evaluation, Cervical vertebrae, Artificial intelligence

Received July 1, 2021; Revised September 25, 2021; Accepted October 1, 2021.

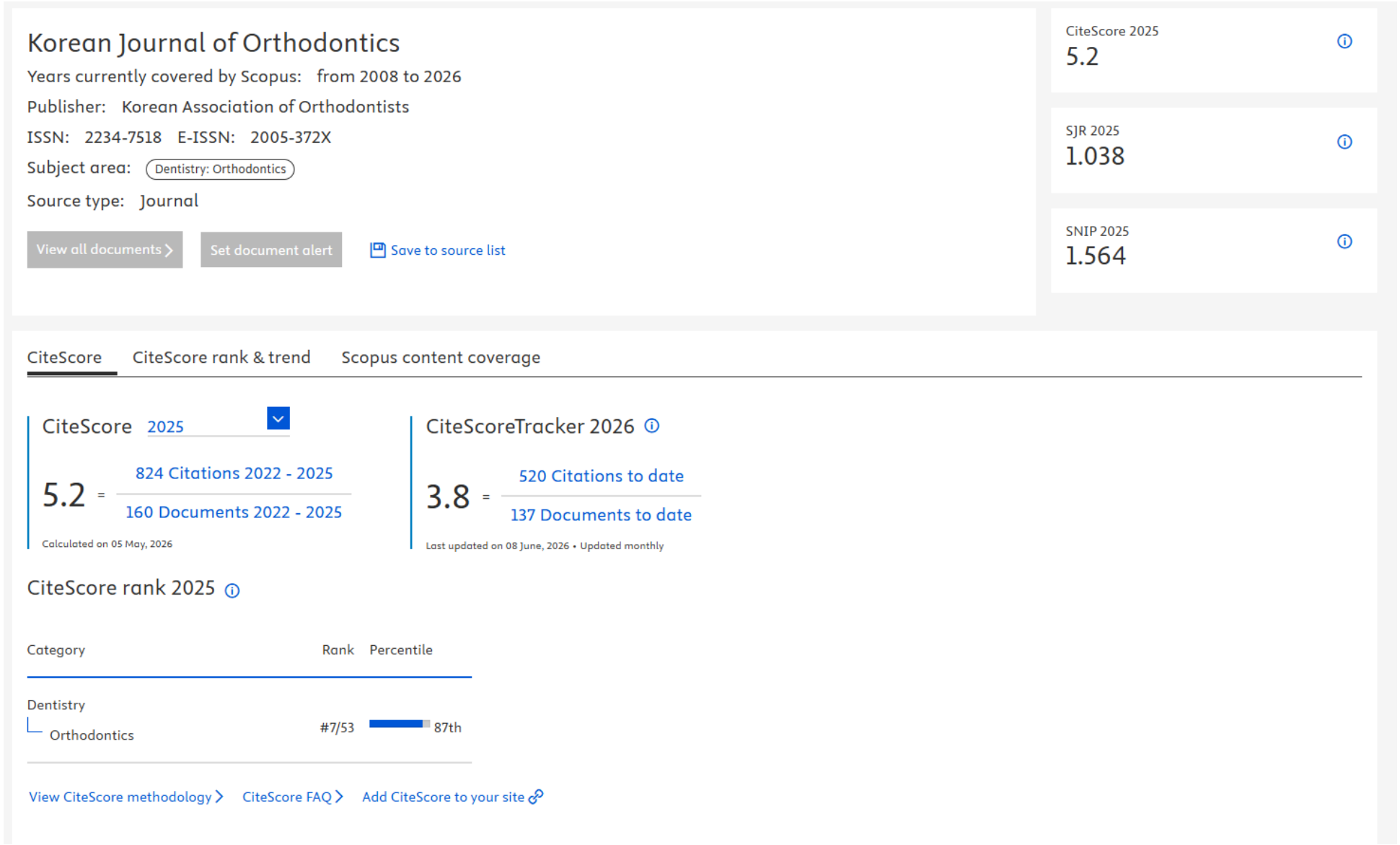
Corresponding author: Mohammad Hossein Rohban.
Assistant Professor, Department of Computer Engineering, Sharif University of Technology, Azadi Ave., Tehran 11155/1639, Iran.
Tel +98-21-66165786 e-mail rohban@sharif.edu

How to cite this article: Mohammad-Rahimi H, Motamadian SR, Nadimi M, Hassanzadeh-Saman S, S. Minabi MA, Mahmoudinia E, Y. Lee V, Rohban MH. Deep learning for the classification of cervical maturation degree and pubertal growth spurts: A pilot study. Korean J Orthod 2022;52:112-122.

© 2022 The Korean Association of Orthodontists.

This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (http://creativecommons.org/licenses/by-nc/4.0) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

- https://www.scopus.com/sourceid/19700168907#tabs=0



Korean Journal of Orthodontics

Years currently covered by Scopus: from 2008 to 2026

Publisher: Korean Association of Orthodontists

ISSN: 2234-7518 E-ISSN: 2005-372X

Subject area: Dentistry: Orthodontics

Source type: Journal

View all documents > | Set document alert | Save to source list

CiteScore 2025
5.2

SJR 2025
1.038

SNIP 2025
1.564

CiteScore | CiteScore rank & trend | Scopus content coverage

CiteScore 2025

5.2 = 824 Citations 2022 - 2025 / 160 Documents 2022 - 2025

Calculated on 05 May, 2026

CiteScoreTracker 2026

3.8 = 520 Citations to date / 137 Documents to date

Last updated on 08 June, 2026 • Updated monthly

CiteScore rank 2025

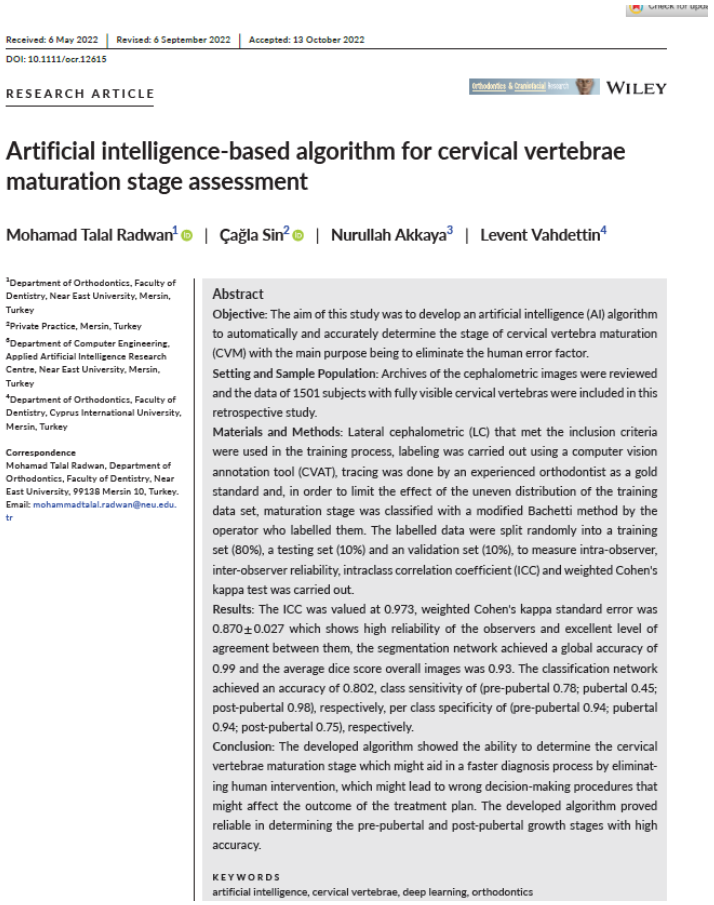
Category	Rank	Percentile
Dentistry Orthodontics	#7/53	87th

View CiteScore methodology > | CiteScore FAQ > | Add CiteScore to your site

Artificial intelligence-based algorithm for cervical vertebrae maturation stage assessment

- Year: 2023
- Journal Name: Orthodontics & Craniofacial Research
- Scopus Quartile: Q1 Percentile: 83rd, 82nd, 80th, 80th

- https://doi.org/10.1111/ocr.12615



Received: 6 May 2022 | Revised: 6 September 2022 | Accepted: 13 October 2022
DOI: 10.1111/ocr.12615

RESEARCH ARTICLE

Orthodontics & Craniofacial Research WILEY

Artificial intelligence-based algorithm for cervical vertebrae maturation stage assessment

Mohamad Talal Radwan[1] | Çağla Sin[2] | Nurullah Akkaya[3] | Levent Vahdettin[4]

[1]Department of Orthodontics, Faculty of Dentistry, Near East University, Mersin, Turkey
[2]Private Practice, Mersin, Turkey
[3]Department of Computer Engineering, Applied Artificial Intelligence Research Centre, Near East University, Mersin, Turkey
[4]Department of Orthodontics, Faculty of Dentistry, Cyprus International University, Mersin, Turkey

Correspondence
Mohamad Talal Radwan, Department of Orthodontics, Faculty of Dentistry, Near East University, 99138 Mersin 10, Turkey.
Email: mohammadtalal.radwan@neu.edu.tr

Abstract

Objective: The aim of this study was to develop an artificial intelligence (AI) algorithm to automatically and accurately determine the stage of cervical vertebra maturation (CVM) with the main purpose being to eliminate the human error factor.

Setting and Sample Population: Archives of the cephalometric images were reviewed and the data of 1501 subjects with fully visible cervical vertebras were included in this retrospective study.

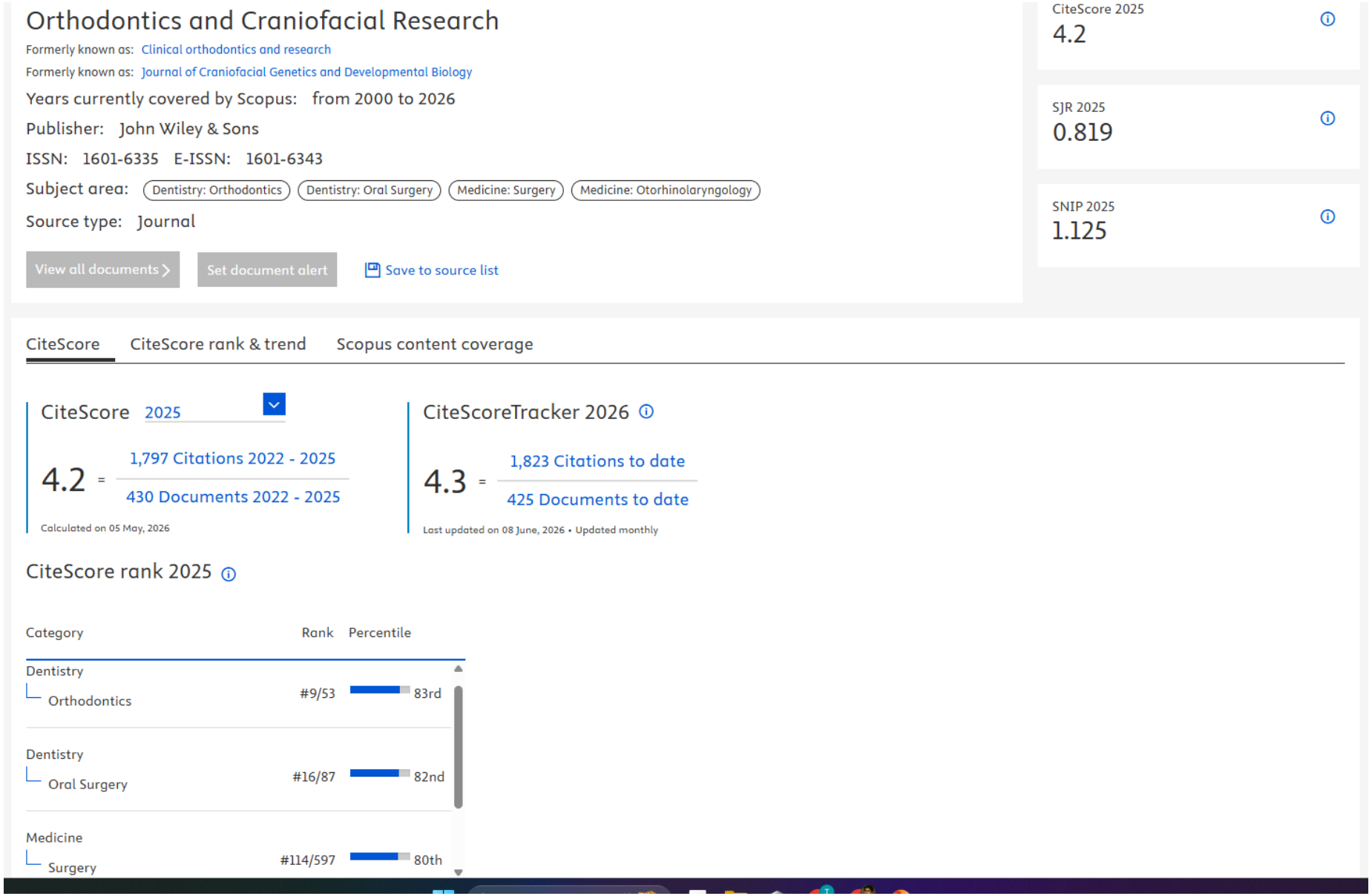
Materials and Methods: Lateral cephalometric (LC) that met the inclusion criteria were used in the training process, labeling was carried out using a computer vision annotation tool (CVAT), tracing was done by an experienced orthodontist as a gold standard and, in order to limit the effect of the uneven distribution of the training data set, maturation stage was classified with a modified Bachetti method by the operator who labelled them. The labelled data were split randomly into a training set (80%), a testing set (10%) and an validation set (10%), to measure intra-observer, inter-observer reliability, intraclass correlation coefficient (ICC) and weighted Cohen's kappa test was carried out.

Results: The ICC was valued at 0.973, weighted Cohen's kappa standard error was 0.870±0.027 which shows high reliability of the observers and excellent level of agreement between them, the segmentation network achieved a global accuracy of 0.99 and the average dice score overall images was 0.93. The classification network achieved an accuracy of 0.802, class sensitivity of (pre-pubertal 0.78; pubertal 0.45; post-pubertal 0.98), respectively, per class specificity of (pre-pubertal 0.94; pubertal 0.94; post-pubertal 0.75), respectively.

Conclusion: The developed algorithm showed the ability to determine the cervical vertebrae maturation stage which might aid in a faster diagnosis process by eliminating human intervention, which might lead to wrong decision-making procedures that might affect the outcome of the treatment plan. The developed algorithm proved reliable in determining the pre-pubertal and post-pubertal growth stages with high accuracy.

KEYWORDS
artificial intelligence, cervical vertebrae, deep learning, orthodontics

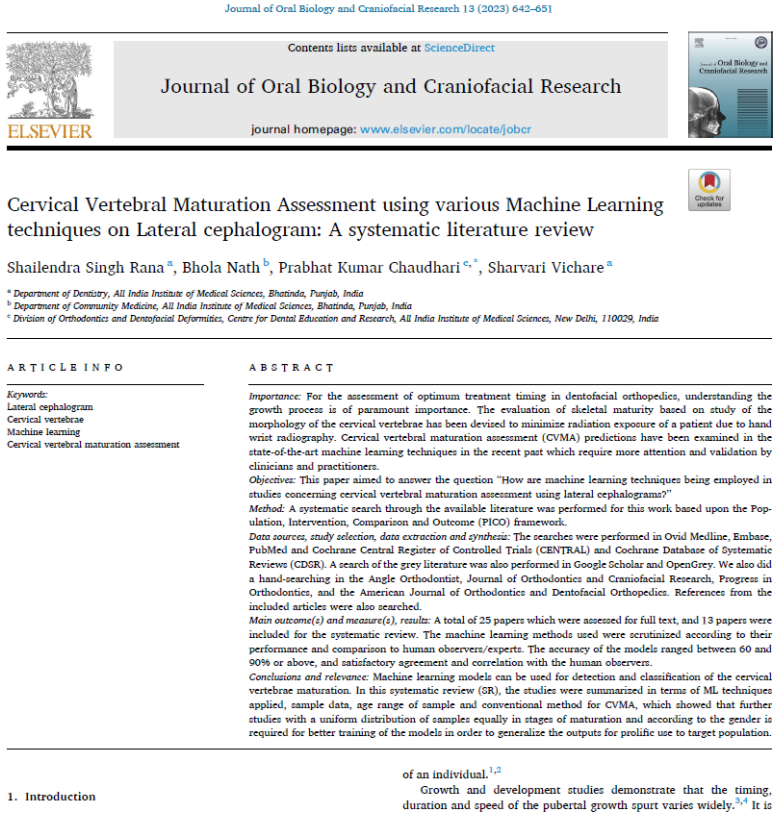
- https://www.scopus.com/sourceid/22848



Cervical Vertebral Maturation Assessment using various Machine Learning techniques on Lateral cephalogram: A systematic literature review

- Year: 2023
- Journal Name: Journal of Oral Biology and Craniofacial Research
- Scopus Quartile: Q1, Q2 Percentile: 79th, 72nd

- https://doi.org/10.1016/j.jobcr.2023.08.005



Journal of Oral Biology and Craniofacial Research 13 (2023) 642–651

Contents lists available at ScienceDirect

Journal of Oral Biology and Craniofacial Research

journal homepage: www.elsevier.com/locate/jobcr

Cervical Vertebral Maturation Assessment using various Machine Learning techniques on Lateral cephalogram: A systematic literature review

Shailendra Singh Rana [a], Bhola Nath [b], Prabhat Kumar Chaudhari [c,*], Sharvari Vichare [a]

[a] Department of Dentistry, All India Institute of Medical Sciences, Bhatinda, Punjab, India
[b] Department of Community Medicine, All India Institute of Medical Sciences, Bhatinda, Punjab, India
[c] Division of Orthodontics and Dentofacial Deformities, Centre for Dental Education and Research, All India Institute of Medical Sciences, New Delhi, 110029, India

ARTICLE INFO

Keywords:
Lateral cephalogram
Cervical vertebrae
Machine learning
Cervical vertebral maturation assessment

ABSTRACT

Importance: For the assessment of optimum treatment timing in dentofacial orthopedics, understanding the growth process is of paramount importance. The evaluation of skeletal maturity based on study of the morphology of the cervical vertebrae has been devised to minimize radiation exposure of a patient due to hand wrist radiography. Cervical vertebral maturation assessment (CVMA) predictions have been examined in the state-of-the-art machine learning techniques in the recent past which require more attention and validation by clinicians and practitioners.

Objectives: This paper aimed to answer the question "How are machine learning techniques being employed in studies concerning cervical vertebral maturation assessment using lateral cephalograms?"

Method: A systematic search through the available literature was performed for this work based upon the Population, Intervention, Comparison and Outcome (PICO) framework.

Data sources, study selection, data extraction and synthesis: The searches were performed in Ovid Medline, Embase, PubMed and Cochrane Central Register of Controlled Trials (CENTRAL) and Cochrane Database of Systematic Reviews (CDSR). A search of the grey literature was also performed in Google Scholar and OpenGrey. We also did a hand-searching in the Angle Orthodontist, Journal of Orthodontics and Craniofacial Research, Progress in Orthodontics, and the American Journal of Orthodontics and Dentofacial Orthopedics. References from the included articles were also searched.

Main outcome(s) and measure(s), results: A total of 25 papers which were assessed for full text, and 13 papers were included for the systematic review. The machine learning methods used were scrutinized according to their performance and comparison to human observers/experts. The accuracy of the models ranged between 60 and 90% or above, and satisfactory agreement and correlation with the human observers.

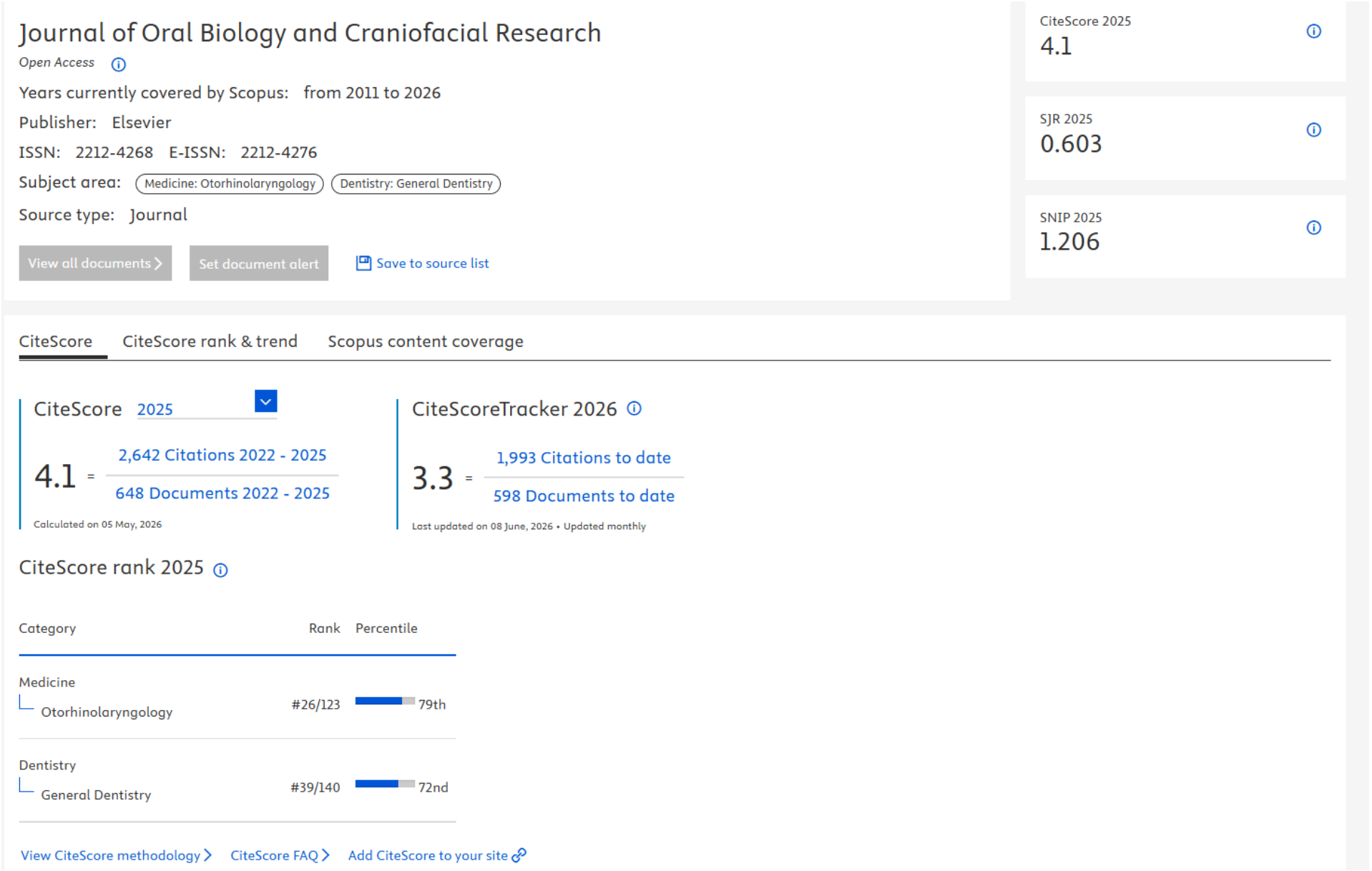
Conclusions and relevance: Machine learning models can be used for detection and classification of the cervical vertebrae maturation. In this systematic review (SR), the studies were summarized in terms of ML techniques applied, sample data, age range of sample and conventional method for CVMA, which showed that further studies with a uniform distribution of samples equally in stages of maturation and according to the gender is required for better training of the models in order to generalize the outputs for prolific use to target population.

1. Introduction

of an individual.[1,2]

Growth and development studies demonstrate that the timing, duration and speed of the pubertal growth spurt varies widely.[3,4] It is

- https://www.scopus.com/sourceid/21100461912



Journal of Oral Biology and Craniofacial Research

Open Access

Years currently covered by Scopus: from 2011 to 2026

Publisher: Elsevier

ISSN: 2212-4268 E-ISSN: 2212-4276

Subject area: Medicine: Otorhinolaryngology | Dentistry: General Dentistry

Source type: Journal

View all documents > | Set document alert | Save to source list

CiteScore 2025
4.1

SJR 2025
0.603

SNIP 2025
1.206

CiteScore | CiteScore rank & trend | Scopus content coverage

CiteScore 2025

4.1 = 2,642 Citations 2022 - 2025 / 648 Documents 2022 - 2025

Calculated on 05 May, 2026

CiteScoreTracker 2026

3.3 = 1,993 Citations to date / 598 Documents to date

Last updated on 08 June, 2026 • Updated monthly

CiteScore rank 2025

Category	Rank	Percentile
Medicine Otorhinolaryngology	#26/123	79th
Dentistry General Dentistry	#39/140	72nd

View CiteScore methodology > | CiteScore FAQ > | Add CiteScore to your site

Accuracy of Artificial Intelligence for Cervical Vertebral Maturation: A Systematic Review

- Year: 2024
- Journal Name: Journal of Clinical Medicine
- Scopus Quartile: Q1 Percentile: 86th
- https://doi.org/10.3390/jcm13144047



Journal of Clinical Medicine

MDPI

Systematic Review

Accuracy of Artificial Intelligence for Cervical Vertebral Maturation Assessment—A Systematic Review

Wojciech Kazimierczak [1,2,*], Maciej Jedliński [3], Julien Issa [4], Natalia Kazimierczak [2], Joanna Janiszewska-Olszowska [3], Marta Dyszkiewicz-Konwińska [4], Ingrid Różyło-Kalinowska [5], Zbigniew Serafin [1] and Kaan Orhan [6,7,8]

1 Department of Radiology and Diagnostic Imaging, Collegium Medicum, Nicolaus Copernicus University in Torun, Jagiellońska 13-15, 85-067 Bydgoszcz, Poland
2 Kazimierczak Private Medical Practice, Dworcowa 13/ u6a, 85-009 Bydgoszcz, Poland
3 Department of Interdisciplinary Dentistry, Pomeranian Medical University in Szczecin, 70-111 Szczecin, Poland
4 Chair of Practical Clinical Dentistry, Department of Diagnostics, Poznań University of Medical Sciences, 61-701 Poznań, Poland
5 Department of Dental and Maxillofacial Radiodiagnostics, Medical University of Lublin, 20-093 Lublin, Poland
6 Department of Dentomaxillofacial Radiology, Faculty of Dentistry, Ankara University, Ankara 06500, Turkey
7 Medical Design Application and Research Center (MEDITAM), Ankara University, Ankara 06500, Turkey
8 Department of Oral Diagnostics, Faculty of Dentistry, Semmelweis University, 1088 Budapest, Hungary
* Correspondence: wojtek.kazimierczak@gmail.com; Tel.: +48-606670881

Abstract: Background/Objectives: To systematically review and summarize the existing scientific evidence on the diagnostic performance of artificial intelligence (AI) in assessing cervical vertebral maturation (CVM). This review aimed to evaluate the accuracy and reliability of AI algorithms in comparison to those of experienced clinicians. **Methods:** Comprehensive searches were conducted across multiple databases, including PubMed, Scopus, Web of Science, and Embase, using a combination of Boolean operators and MeSH terms. The inclusion criteria were cross-sectional studies with neural network research, reporting diagnostic accuracy, and involving human subjects. Data extraction and quality assessment were performed independently by two reviewers, with a third reviewer resolving any disagreements. The Quality Assessment of Diagnostic Accuracy Studies (QUADAS)-2 tool was used for bias assessment. **Results:** Eighteen studies met the inclusion criteria, predominantly employing supervised learning techniques, especially convolutional neural networks (CNNs). The diagnostic accuracy of AI models for CVM assessment varied widely, ranging from 57% to 95%. The factors influencing accuracy included the type of AI model, training data, and study methods. Geographic concentration and variability in the experience of radiograph readers also impacted the results. **Conclusions:** AI has considerable potential for enhancing the accuracy and reliability of CVM assessments in orthodontics. However, the variability in AI performance and the limited number of high-quality studies suggest the need for further research.

Keywords: artificial intelligence (AI); lateral cephalogram; cervical vertebrae; machine learning; cervical vertebral maturation assessment; skeletal maturity

check for updates

Citation: Kazimierczak, W.; Jedliński, M.; Issa, J.; Kazimierczak, N.; Janiszewska-Olszowska, J.; Dyszkiewicz-Konwińska, M.; Różyło-Kalinowska, I.; Serafin, Z.; Orhan, K. Accuracy of Artificial Intelligence for Cervical Vertebral Maturation Assessment—A Systematic Review. *J. Clin. Med.* **2024**, *13*, 4047. https://doi.org/10.3390/jcm13144047

Academic Editors: Vincenzo Grassia, Letizia Perillo and Fabrizia d'Apuzzo

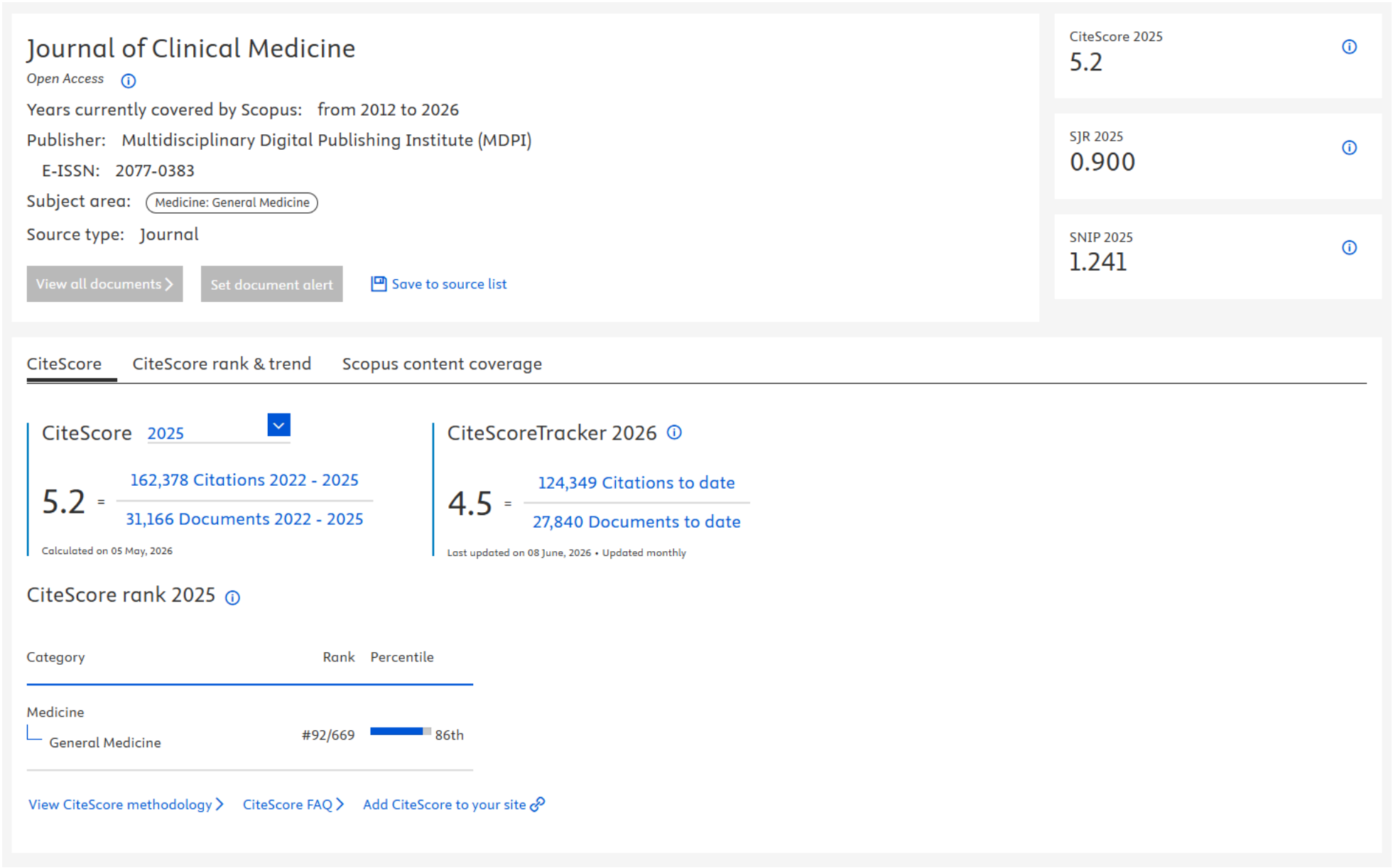
Received: 4 June 2024
Revised: 3 July 2024
Accepted: 8 July 2024
Published: 10 July 2024

Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

In the last few years, there has been an increase in the amount of scientific evidence supporting the diagnostic accuracy and effectiveness of AI in various clinical scenarios [1]. Due to the nature of diagnostic imaging and its repetitive analysis of specific image features, radiology is an area of medicine in which AI is developing most rapidly [2]. Owing to its significant use of imaging and emphasis on cephalometric analysis, orthodontics is particularly well suited for the implementation of AI [3]. Recently, the effectiveness of AI has been

- https://www.scopus.com/sourceid/21101054449



Journal of Clinical Medicine

Open Access

Years currently covered by Scopus: from 2012 to 2026

Publisher: Multidisciplinary Digital Publishing Institute (MDPI)

E-ISSN: 2077-0383

Subject area: Medicine: General Medicine

Source type: Journal

View all documents > | Set document alert | Save to source list

CiteScore 2025
5.2

SJR 2025
0.900

SNIP 2025
1.241

CiteScore | CiteScore rank & trend | Scopus content coverage

CiteScore 2025

5.2 = 162,378 Citations 2022 - 2025 / 31,166 Documents 2022 - 2025

Calculated on 05 May, 2026

CiteScoreTracker 2026

4.5 = 124,349 Citations to date / 27,840 Documents to date

Last updated on 08 June, 2026 • Updated monthly

CiteScore rank 2025

Category	Rank	Percentile
Medicine		
General Medicine	#92/669	86th

View CiteScore methodology > | CiteScore FAQ > | Add CiteScore to your site